

The background is a solid blue gradient. At the top, there are several wavy, horizontal lines in shades of light blue and cyan, creating a layered, wave-like effect.

Digital Signature certificate

What is a Digital Signature Certificate?

- Digital Signature Certificates (DSC) are the digital equivalent of physical or paper certificates.
- Digital Signatures make use of the public key encryptions to create the signatures.
- A digital signature certificate (DSC) contains information about the user's name, pin code, country, email address, date of issuance of certificate and name of the certifying authority.



Paper signatures v/s Digital Signatures



V/s

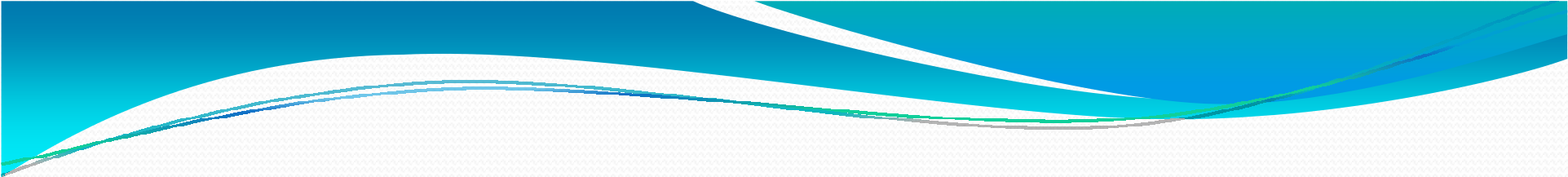


Parameter	Paper	Electronic
Authenticity	May be forged	Can not be copied
Integrity	Signature independent of the document	Signature depends on the contents of the document
Non-repudiation	a. Handwriting expert needed b. Error prone	a. Any computer user b. Error free

*Digital Certificates



- *Digital Certificate is a data with digital signature from one trusted Certification Authority (CA).
- *This data contains:
 - *Who owns this certificate
 - *Who signed this certificate
 - *The expired date
 - *User name & email address



Different Types of Signatures

- **1. Wet Signature**
- A wet signature is a traditional handwritten signature made with pen and ink.
- **2. Electronic Signatures**
- Electronic signatures, or eSignatures, refer to digital marks that represent a person's agreement or approval. These could include a scanned signature, a typed name, or a stylus-drawn signature on a device.

- **3. Digital Signatures**

- Digital signatures, also known as PKCS7 encrypted signatures, are equipped with encryption technology to add authenticity. People can use these signatures by certifying authorities who encrypt the public and private keys for the user.

- **4. Aadhaar eSign**

- Users add their Aadhaar credentials to sign their documents. The credentials are validated at the Certifying authority's gateway, and an encrypted

- **5. Initials, Symbols, and Marks**

- In certain scenarios, a signature is a unique marking, initials, or just a symbol instead of a full name.

Why is Digital Signature Certificate (DSC) required?



- Physical documents are signed manually, similarly, electronic documents, for example e-forms are required to be signed digitally using a Digital Signature Certificate.



Who issues DSC in India?

- In India, Digital Signature Certificates (DSCs) are issued by licensed Certifying Authorities (CAs). These CAs are authorized by the Controller of Certifying Authorities (CCA), a government body under the Ministry of Electronics and Information Technology. The CCA regulates and oversees the activities of these CAs.

Type of Certificates



Sign

Sign DSC can only be used for signing documents. Signing via DSC gives the assurance of not only the integrity of the signer but also the data. It is proof of untampered and unaltered data.



Encrypt

Encrypt DSC can only be used to encrypt a document, Encrypt DSC is fit for e-commerce documents, legal documentation and sharing documents that are highly confidential and contains information that needs to be protected.

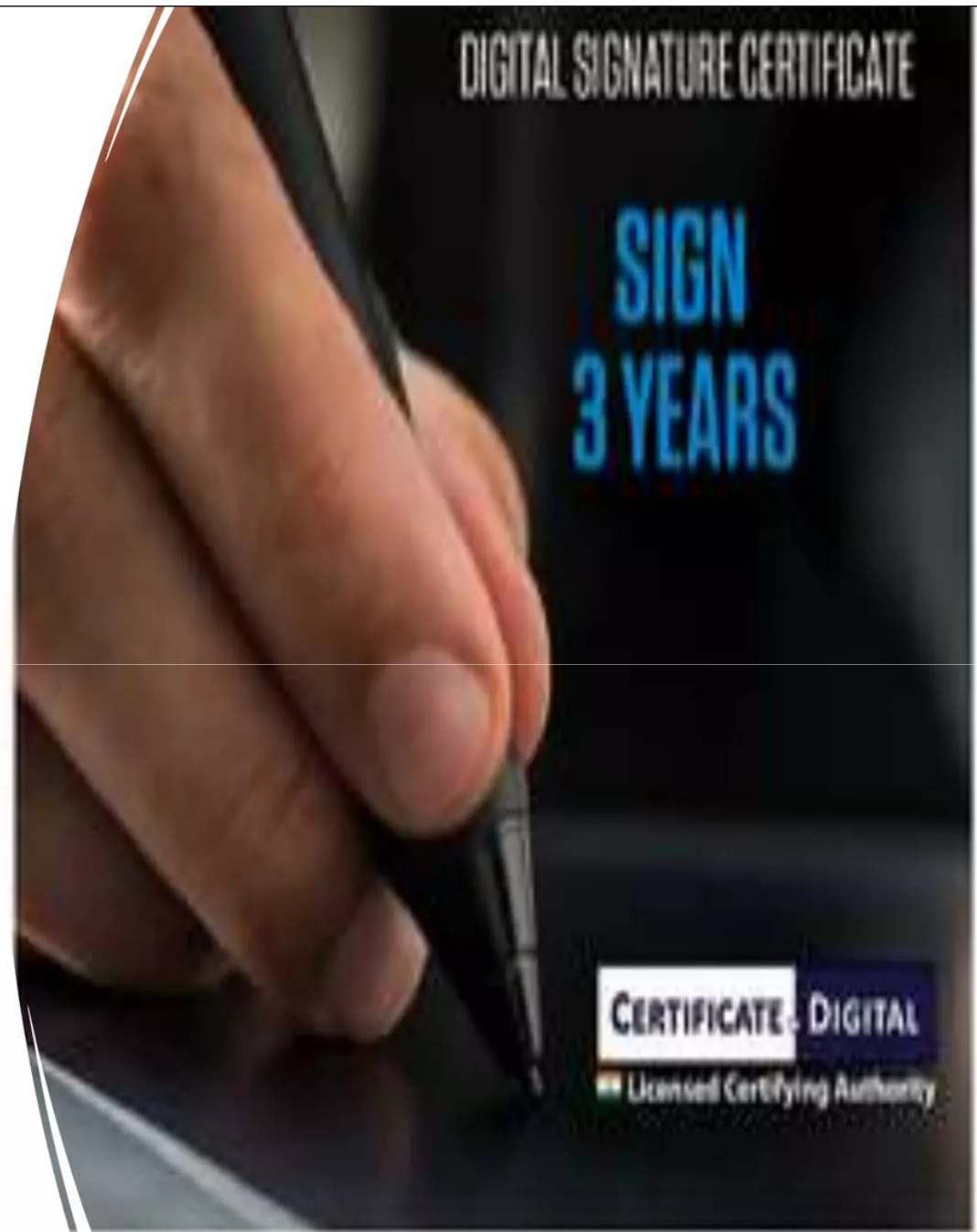


Sign & Encrypt

Sign & Encrypt DSC can be used for both signing and encrypting. It is convenient for users who need to authenticate and maintain the confidentiality of the information shared.

What is the validity period of a Digital Signature Certificate?

- The Certifying Authorities are authorized to issue a Digital Signature Certificate with a validity of one, two or three years.



What is the process of obtaining DSC from Certifying Authority?

- Digital Signature Certificate (DSC)
Applicants can directly approach Certifying Authorities (CAs) with original supporting documents, and self-attested copies will be sufficient in this case
- DSCs can also be obtained, wherever offered by CA, using Aadhar eKYC based authentication, and supporting documents are not required in this case
- A letter/certificate issued by a Bank containing the DSC applicant's information as retained in the Bank database can be accepted. Such letter/certificate should be certified by the Bank Manager .

Classes of DSC

- **Class 1** : These are issued to individual/private subscribers and are used to confirm that the user's name and email contact details from the clearly defined subject lie within the database of the certifying authority.
- **Class 2** : These are issued to the director/signatory authorities of the companies for the purpose of e-filing with the Registrar of Companies (ROC).
- **Class 3** : These certificates are used in online participation/bidding in e-auctions and online tenders anywhere in India.

What are the different classes of Digital Signature Certificates?

- Though there are three classes of DSCs, namely Class 1, Class 2 and Class 3 DSC, the usage of Class 1 and Class 2 DSC is discontinued in India. Class 3 DSC is now mandated to be used for all types of transactions in India.
- Class 3 DSC provides the highest level of assurance and trust issued in the name of individuals as well as organizations. Applicants are required to appear before the certifying authority of the registration authority for obtaining Class 3 DSC. It comes with a validity of one or two or three years.

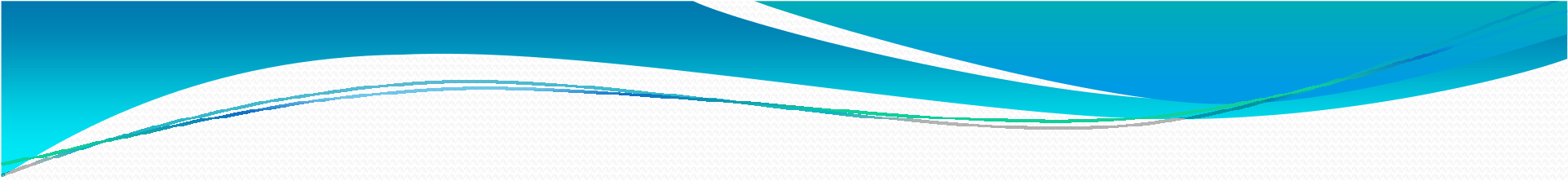
Benefits of a Digital Signature Certificate

Digital Signature Certificates are helpful in authenticating the personal information details of the individual holder when conducting business online.

- Reduced cost and time
- Data integrity
- Authenticity of documents

Cost of DSC

Validity	Sign	Sign and Encrypt
1 Year e-KYC	₹1,000.00 Buy	₹2,000.00 Buy
2 Years e-KYC	₹1,150.00 Buy	₹2,400.00 Buy
3 Years e-KYC	₹1,750.00 Buy	₹3,700.00 Buy



Fro Prevention and Misuse

- Don't Hand over your DSC Media USB/Card to any one
- Don't tell your PIN to anyone
- Document Digitally Signed carries same legal status as manually signing as per the IT Act

Certifying Authorities

1. National Informatics Center (NIC) 
2. IDRBT Certifying Authority 
3. SafeScript CA Services, Sify Communications Ltd. 
4. (n) Code Solutions CA 
5. E-MUDHRA 
6. CDAC 
7. NSDL 
8. Capricorn 
9. Verasys 

Thank You!

